

to form a personal area network (PAN), also known as a piconet. This process is done through discovery, with one device making itself discoverable by the other device.

Bluetooth is a common mobile connectivity method because it has low power consumption requirements and a short-range signal. One of the most popular communication methods is 802.11 wireless. Point-to-point is a one-to-one connection, involving conversation between two devices. In a point-to-multipoint connection, multiple devices can connect but this doesn't necessarily mean that you can stream media from two devices at a time.

Common security concerns: Some of the most vital security concerns for Bluetooth technology are:

- **Bluejacking:** Sending of unsolicited messages to another device via Bluetooth
- **Bluesnarfing:** Unauthorised access to a Bluetooth-enabled device to steal data from it

Security measures: The security measures that are most effective against these security threats are:

- Update with latest patches
- Monitoring
- Encryption

RFID

Radio frequency identification (RFID) refers to a wireless system comprising two components — tags and readers. The reader is a device that has one or more antennas that emit radio waves and receive signals back from the RFID tag.

There are two types of RFID tags — active and passive tags. An active tag can broadcast a signal over a larger distance because it contains a power source. A passive tag, on the other hand, isn't powered but is activated by a signal sent from the reader.

RFID uses radar technology, and radio energy is transmitted for bi-directional communication.

Common security concerns: Major security concerns hounding RFID

technology include:

- Data capture
- Spook reader
- Signal jamming
- Decrypt communication

Security measures: The measures having the most impact on RFID security concerns are:

- Cryptography
- Blocker tags; these prevent unauthorised readers

RFID is commonly used as a geofencing security measure to define a geographic perimeter. When the device enters or exits the perimeter, alerts are sent through this technology.

NFC

Near-field communication (NFC) is a set of standards for contactless communication between devices. NFC chips in mobile devices generate electromagnetic fields. This allows a device to communicate with other devices. NFC is an extension of RFID technology.

Near-field communication transmits data through electromagnetic radio fields to enable two devices to communicate with each other. To work, both devices must contain NFC chips, as transactions take place within a very short distance. NFC-enabled devices must be either physically touching or within a few centimetres of each other for data transfer to occur.

NFC began in the payment-card industry and is evolving to include applications in numerous industries worldwide.

The NFC standard has three modes of operation.

- **Peer-to-peer mode:** Information shared is exchanged between two mobile devices directly.
- **Read/write mode:** An active device receives data from a passive device.
- **Card emulation:** The device is used as a contactless credit card.

It emulates a payment card or other physical card in card readers, magnetic-stripe readers, and contactless card readers used to make payments directly

from your mobile device.

Common security concerns: The most relevant security concerns for NFC technology are:

- Remote capture
- Frequency jamming
- On-path attacks
- Loss of NFC device control -- digital pick-pocketing

Security measures: The security measures that most effectively handle NFC security concerns are:

- Encryption
- Up-to-date patching at all times
- Turning it off when not in use

Mobile networks

Mobile networking has evolved significantly since the introduction of the first-generation (1G) mobile network in the 1980s. G refers to generation. Each generation is defined as a set of telephone network standards, which detail the technological implementation of a particular mobile phone system.

1G: 1G was the first generation of wireless cellular technology. It was analogue and supported voice calls only. The maximum speed of 1G was 2.4Kbps.

2G: 2G introduced digital technology and the ability to send SMS (short message service) and plain text-based messages. GSM and CDMA were also introduced during this period. The maximum speed of 2G with general packet radio service (GPRS) was 50Kbps. The maximum theoretical speed was 384Kbps with enhanced data rates for GSM evolution (EDGE).

Before making the major leap from 2G to 3G wireless networks, the lesser-known 2.5G and 2.75G were interim standards that bridged the gap.

3G: 3G enabled web browsing, email, video downloading, picture sharing, and so on. The maximum speed of 3G was around 2Mbps for non-moving devices and 384Kbps for devices in moving vehicles.

4G: 4G applications include amended mobile web access, IP telephony, gaming services,